BST Physics Curriculum Vol 5 Chapter 11 — POVMs + Quantum Information Basics v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

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Vol 5 Chapter 11 — POVMs + Quantum Information Basics

Chapter motivation

Standard QM measurement framework extends to POVMs (Positive Operator-Valued Measures) — generalized measurements with multiple outcomes per detector. Quantum information theory (Nielsen-Chuang) builds on POVMs + entanglement + Bell-state operations. BST derives POVMs from substrate via T2479 (Saturday) extending K67 Born = Bergman RATIFIED. Quantum information primitives substrate-cartography.

Section 11.0b — Reader-grade 3-level pedagogy

Level 1: Standard QM POVMs derive from substrate via T2479 (Saturday SP-31 #283) extending K67 Born = Bergman RATIFIED foundation; quantum information primitives (no-cloning + entanglement + teleportation + quantum gates) inherit from substrate K-type operator algebra (Vol 1 Ch 6).

Level 2 (graduate): T2479 POVM substrate-derivation (Saturday): for POVM {E_i}*{i ∈ I} on observer's K-type coverage, P(outcome i) = ⟨ψ|E_i|ψ⟩ with E_i =* Σ{K-types ∈ B_i} K_B · proj_K positive operator on Bergman H^2(D_IV^5); Bergman positive-definiteness ensures POVM positivity; T2417 4-Zone observer bandwidth ensures Σ_i E_i = I completeness. Standard POVM machinery (Naimark dilation, Kraus operators, Lindblad master equation) substrate-cartography readings. Quantum information primitives: no-cloning theorem (Wootters-Zurek 1982) follows from linearity of substrate K-type evolution; entanglement = K-type tensor product non-separability on Bergman H^2(D_IV^5)^⊗N; Bell states + quantum teleportation = K-type basis transformations on multi-particle substrate states; quantum gates (Hadamard, CNOT, T) = unitary K-type rotations on observer Hilbert space. Sub-Tsirelson 1/8 = 1/2^N_c (Vol 5 Ch 8) provides BST distinction from standard QI; substrate-mediation signature testable at Bell-experiment level. Quantum error correction codes substrate-cartography: stabilizer codes inherit Pin(2) Z_2 grading structure (T1925 rank=2); CSS codes substrate-derivation pending multi-month.

Level 3 (5th-grader): Standard quantum information theory (the math behind quantum computers) builds on POVMs (generalized quantum measurements), entanglement (correlated quantum states), and quantum gates (basic operations on qubits). BST derives all this from the substrate D_IV^5: POVMs come from T2479 (proven Saturday); entanglement = correlated K-type modes on the substrate; quantum gates = unitary rotations of substrate K-type states. The famous "no-cloning theorem" follows from the linearity of substrate K-type evolution. Quantum computing primitives (Hadamard gate, CNOT, T gate) all inherit from substrate. BST adds: sub-Tsirelson 1/8 deviation is the substrate-mediation signature — Bell experiments at high precision can test this.

Section 11.1 — Standard QM POVMs

POVM {E_i}_{i ∈ I}: positive operators with Σ_i E_i = I; P(outcome i) = Tr(ρ E_i) for state ρ. Generalizes projective measurements; includes Naimark dilation (POVM = projective on enlarged Hilbert space).

Section 11.2 — T2479 Substrate-Derivation (Saturday SP-31 #283)

Per T2479: E_i = Σ_{K-types ∈ B_i} K_B · proj_K on Bergman H^2(D_IV^5); positive operator by Bergman positive-definiteness; Σ_i E_i = I by T2417 4-Zone observer bandwidth.

Standard POVM machinery substrate-cartography readings.

Section 11.3 — Quantum Information Primitives

- No-cloning theorem (Wootters-Zurek 1982): linearity of substrate K-type evolution
- Entanglement: K-type tensor product non-separability on Bergman H^2(D_IV^5)^⊗N
- Bell states + teleportation: K-type basis transformations on multi-particle substrate states
- Quantum gates (H, CNOT, T): unitary K-type rotations on observer Hilbert space

Section 11.4 — Sub-Tsirelson 1/8 BST Distinction (Vol 5 Ch 8 Cross-Link)

Standard quantum information uses Tsirelson saturation as benchmark; BST predicts substrate-mediation sub-Tsirelson 1/8 = 1/2^N_c (Vol 5 Ch 8). Operationally distinguishable at Bell experiment design $300-500K (SP-30).

Section 11.5 — Quantum Error Correction Substrate-Cartography

Stabilizer codes inherit Pin(2) Z_2 grading structure (T1925 rank=2; Paper #133 spin-statistics). CSS codes substrate-derivation pending multi-month (Vol 7 Information Theory cross-link).

Section 11.6 — Honest scope

- POVMs derived from substrate ⍰ (T2479 SP-31 #283 closure)
- Quantum information primitives substrate-cartography ⍰
- Sub-Tsirelson 1/8 BST distinction (Bell falsifier Vol 5 Ch 8)
- Quantum error correction stabilizer-code substrate-derivation pending multi-month

Section 11.7 — Connection to other chapters

- Vol 5 Ch 7 Born Rule + Measurement (K67 + T2479)
- Vol 5 Ch 8 Bell-CHSH + sub-Tsirelson 1/8
- Vol 5 Ch 10 Decoherence (substrate-coupled environments)
- Vol 7 Information Theory (full quantum information textbook cross-link)

— Lyra, Vol 5 Ch 11 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT